

APPENDIX

How to read the figures. — given --- auxiliary (added in the proof)
— what is being proved or asked

Opening paragraph (book p. 275): the Appendix collects all the postulates for plane geometry together with a selection of the definitions and theorems. The authors recommend reading it through rather than merely consulting it, on the grounds that the order in which the theorems are arranged is itself a picture of how Euclidean geometry is built up. Summarised, not transcribed — see `appendix-UNCERTAIN.md`, item 1.

THE POSTULATES

There are three groups of postulates, and three postulates listed singly. They are:

- Postulates of incidence (3 postulates)
- Postulates of order or betweenness (5 postulates)
- Postulates of congruence (7 postulates)
- The postulate of Archimedes
- The completeness postulate
- The parallel postulate.

The Incidence Postulates

1. There are at least three points not all on a line.
2. For any two different points, there is exactly one line containing these points.
3. Every line contains at least two points.

The Betweenness Postulates

4. If B is between A and C , then A , B , and C are three different, or distinct, points on a line.
5. Of any three points on a line, exactly one is between the other two.
6. Four points on a line can be named A_1, A_2, A_3, A_4 so that the betweenness relations among them are exactly those given by the order of the subscripts.
7. If A and B are two points, then there is at least one point C such that B is between A and C , and at least one point D such that D is between A and B .
8. Every line l separates the plane: the points not on l fall into two sets, the two *sides* of l , such that
 - (a) two points of the same set have no point of l between them;
 - (b) two points of different sets have a point of l between them.

Linear Congruence Postulates

9. Given A, B on a line l and A' on a line l' , there is exactly one point B' on a given side of A' with $AB \cong A'B'$. Order is immaterial: $AB \cong A'B'$ also gives $AB \cong B'A'$, $BA \cong A'B'$ and $BA \cong B'A'$.

10. Segment congruence is reflexive, symmetric and transitive:
- (a) $AB \cong AB$.
 - (b) If $AB \cong A'B'$, then $A'B' \cong AB$.
 - (c) If $AB \cong A'B'$ and $A'B' \cong A''B''$, then $AB \cong A''B''$.
11. Segments may be added and subtracted. Let B lie between A and C on a line l , and B' between A' and C' on a line l' .
- (a) If $AB \cong A'B'$ and $BC \cong B'C'$, then $AC \cong A'C'$.
 - (b) If $AC \cong A'C'$ and $BC \cong B'C'$, then $AB \cong A'B'$.

Archimedes' Postulate

12. Given a segment AB on a line l and any other segment CD , finitely many points $A_1, A_2, \dots, A_n$ of l can be laid off in order from A with every segment $AA_1, A_1A_2, \dots, A_{n-1}A_n$ congruent to CD , and with either $B = A_n$ or B between A_{n-1} and A_n .

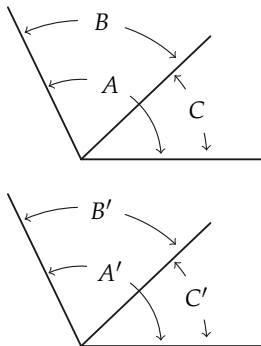
The Completeness Postulate

13. For every line l , every point A on l and every positive real number x , there is a point B on a given side of A with $\overline{AB} = x$.

Congruence Postulates for Angles

14. If $\angle A$ is not a straight angle and r' is a ray from A' on a line l' , then on a given side of l' there is exactly one ray s' from A' making the angle at A' with sides r', s' congruent to $\angle A$; we write $\angle A' \cong \angle A$. If $\angle A$ is a straight angle, the straight angle at A' with side r' is the only such angle.

15. Angle congruence is reflexive, symmetric and transitive:
- (a) $\angle A \cong \angle A$.
 - (b) If $\angle A \cong \angle B$, then $\angle B \cong \angle A$.
 - (c) If $\angle A \cong \angle B$ and $\angle B \cong \angle C$, then $\angle A \cong \angle C$.
16. In the figure, if any two of the three pairs $\angle A$ and $\angle A'$, $\angle B$ and $\angle B'$, $\angle C$ and $\angle C'$ are congruent, then so is the third pair.
That is:
- (a) If $\angle A \cong \angle A'$ and $\angle B \cong \angle B'$, then $\angle C \cong \angle C'$.
 - (b) If $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$, then $\angle A \cong \angle A'$.
 - (c) If $\angle C \cong \angle C'$ and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$.
- The angle A may be a straight angle.



17. If for $\triangle ABC$ and $\triangle A'B'C'$, $AB \cong A'B'$, $AC \cong A'C'$, and $\angle A \cong \angle A'$, then $\angle B \cong \angle B'$ and $\angle C \cong \angle C'$.

The Parallel Postulate

18. Through a point not on a line there is no more than one line parallel to the given line.

SELECTED DEFINITIONS

Headnote (book p. 277): not every definition is equally load-bearing, and the ones gathered here are the essential ones; a few are given in abbreviated form where the short form cannot be misread. Summarised, not transcribed.

1. A *geometric figure* is any set of points in the plane.
2. The *line segment* (or *segment*) determined by two points A, B of a line is the set consisting of A, B and all points between them, written " AB "; A and B are its *ends*.
3. For three points A, B, C not all on a line, the points of the segments AB, BC, AC form a *triangle*, written $\triangle ABC$; A, B, C are its *vertices* and AB, AC, BC its *sides*.
4. For O, A, B on a line, A and B are on the *same side* of O if O is not between them, and on *opposite sides* of O if O is between them.
5. A *ray* from O consists of O together with all points on one side of O of a line through O .
6. An *angle* is the set of all points on two rays from a point O , its *vertex*; the rays are the *sides*. If the rays lie on one line the angle is a *straight angle*.
7. For an angle that is not straight, with vertex O and sides r_1, r_2 on lines l_1, l_2 : its *interior* is the set of points lying both on the same side of l_1 as r_2 and on the same side of l_2 as r_1 ; its *exterior* is the set of points belonging neither to the angle nor to its interior.
8. Two angles are *adjacent* if they share a side and vertex and have no interior point in common.
9. Let AB and $A'B'$ be segments, $A'B'$ on a line l' , and let B'' be the point on the same side of A' as B' with $AB \cong A'B''$. Then $AB > A'B'$ if B' is between A' and B'' , and $AB < A'B'$ if B'' is between A' and B' .

10. For the definition of *length of a segment* read Chapter 5, Section 4.
11. A *right* angle is one congruent to its supplement.
12. Two triangles are *congruent* if they can be labeled $\triangle ABC$ and $\triangle A'B'C'$ in such a way that all three pairs of corresponding sides and all three pairs of corresponding angles are congruent. This is written $\triangle ABC \cong \triangle A'B'C'$.
13. Let $\angle A$ and $\angle B$ be angles, neither of them straight. Copy $\angle B$ as $\angle B'$ against the first side of $\angle A$, placing it on the same side of that first side as the second side of $\angle A$. Then $\angle B < \angle A$ if the second side of $\angle B'$ is interior to $\angle A$, and $\angle B > \angle A$ if it is exterior. If $\angle A$ is straight and $\angle B$ is not, then $\angle A > \angle B$.
14. Given n points $P_1, P_2, \dots, P_n$, the points lying on the segments $P_1P_2, P_2P_3, \dots, P_{n-1}P_n$ form a *polygonal path* joining P_1 to P_n ; the points P_i are its *vertices*. If $P_1 = P_n$, the path is called a *polygon*, designated polygon $P_1P_2 \dots P_{n-1}$.
A polygon is *convex* if, for every two of its points, every point between them lies on the polygon or in its interior. A convex polygon that is both equiangular and equilateral is *regular*.
15. Two lines are *parallel* if they do not intersect.
16. If the ray OB is in the interior of $\angle AOC$, or $\angle AOC$ is straight, then $\angle AOC$ is the *sum* of $\angle AOB$ and $\angle BOC$.
17. The *ratio* of segment AB to segment CD is the number $\overline{AB} \div \overline{CD}$. Segments AB, CD are *proportional* to $A'B', C'D'$ when $\overline{AB}/\overline{A'B'} = \overline{CD}/\overline{C'D'}$; the sides of $\triangle ABC$ are *proportional* to those of $\triangle A'B'C'$ when $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CA}/\overline{C'A'}$.
18. A *circle* with center O and positive number r is the set of points A with $\overline{OA} = r$. For A on the circle, OA is a *radius* (plural *radii*); a segment AB with A, B on the circle and O on AB is a *diameter*.
19. For a circle with center O and radius r : its *interior* is the set of points B with $\overline{OB} < r$, its *exterior* the set of points C with

$\overline{OC} > r$. A *chord* is a segment joining two points E, F of the circle, and line EF is a *secant*. A *central angle* has its vertex at the center; an *inscribed angle* has its vertex on the circle and meets the circle again on each of its two sides.

20. If to each positive integer n a number x_n is assigned, the numbers $x_1, x_2, \dots, x_n$ form an *infinite sequence*, written $\{x_n\}$, and x_n is its n th term.
21. L is the *limit of the sequence* x_n if x_n is necessarily arbitrarily close to L for large n ; written $L = \lim\{x_n\}$.
22. The *circumference* C of a circle is the limit of the perimeters p_n of the regular inscribed polygons: $C = \lim\{p_n\}$.
23. The number C/d , the same for every circle, is denoted by the Greek letter π (pi).
24. The *area* S of a circle is the limit of the areas of the inscribed regular polygons: $S = \lim\{S_n\}$.
25. For a circular arc, let l_n be the length of the regular polygonal path of 2^n sides inscribed in it. The *length* l of the arc is $l = \lim\{l_n\}$.
26. On a circle of radius r , an arc of length $\pi r/180$ is subtended by a central angle of *one degree* (1°).
27. If a central angle of a circle of radius r subtends an arc of length L , then the *measure of the angle in degrees* is $(L/\pi r) \cdot 180^\circ$.

SELECTED THEOREMS

Headnote (book p. 280): the theorems below were picked to show how the ideas were developed and, above all, in what order; the list is avowedly incomplete, and where a statement here is looser than the one in the text the reader is asked to supply the missing precision or turn back to the chapter. Summarised, not transcribed.

1. Two different lines intersect in at most one point.
2. A point O of a line l separates l into two *sides* of O , such that for P, Q on the same side either OPQ or OQP , and for P, Q on different sides POQ .
3. If P and Q are both interior to $\angle O$, so is the segment PQ .
4. A ray r from O other than the sides r_1, r_2 of $\angle O$ lies wholly in the interior or wholly in the exterior of $\angle O$.
5. If P is interior to $\angle O$ and Q exterior, the segment PQ contains a point of the angle.
6. For any two segments exactly one of $AB > A'B', AB \cong A'B', AB < A'B'$ holds.
7. If $AB > CD$ and $CD \cong EF$, then $AB > EF$.
8. If $AB \cong A'B'$, then $\overline{AB} = \overline{A'B'}$.
9. If $\overline{AB} = \overline{A'B'}$, then $AB \cong A'B'$.
10. $AB > CD$ if and only if $\overline{AB} > \overline{CD}$.
11. The S.A.S. theorem for congruent triangles.
12. The A.S.A. theorem.
13. The S.S.S. theorem.
14. For any two angles exactly one of $\angle A < \angle B, \angle A \cong \angle B, \angle A > \angle B$ is true.
15. All right angles are congruent to one another.
16. An exterior angle of a triangle is greater than either opposite interior angle.
17. In $\triangle ABC$, $AB > AC$ if and only if $\angle C > \angle B$.
18. In $\triangle ABC$, $\overline{AB} + \overline{BC} > \overline{AC}$.
19. If l is any line and A any point not on l , there exists at least one line on A parallel to l .
20. If parallel lines are cut by a transversal, the alternate interior angles are congruent.
21. The sum of the angles of a triangle is a straight angle.

22. If a transversal is cut by three parallel lines in congruent segments, the same is true for every transversal of these lines.
23. If for $\triangle ABC$ and $\triangle A'B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$, then the triangles are similar.
24. If $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'} = \overline{BC}/\overline{B'C'}$, then $\triangle ABC$ is similar to $\triangle A'B'C'$.
25. In a right triangle the altitude on the hypotenuse forms two right triangles similar to the given triangle.
26. In $\triangle ABC$ with $\angle C$ a right angle, $\overline{AC}^2 + \overline{BC}^2 = \overline{AB}^2$.
27. The area of a triangle is one half the base times the altitude.
28. Circles with centers A and B , each of radius $r > \frac{1}{2}\overline{AB}$, intersect in two points on opposite sides of the line $\overline{AB}$.
29. Exactly one circle is circumscribed about a regular polygon.
30. If regular polygons of n sides are inscribed in two circles, then the perimeters of the polygons are proportional to the radii of the circles.
31. For a given circle let p_n be the perimeter of a regular polygon of n sides inscribed in the circle. The numbers p_n , $n = 3, 4, 5, \dots$, form a sequence which has a limit.
32. The ratio C/d (circumference to diameter) is the same for all circles.
33. If S_n is the area of a regular polygon of n sides inscribed in a circle, then limit $\{S_n\}$ exists.
34. The area of a circle is equal to one half the circumference times the radius.
35. If two central angles are congruent, they subtend arcs of equal length.
36. If $\widehat{AB}$ and $\widehat{BC}$ are adjacent arcs (that is, if they have only point B in common), then length $\widehat{ABC} = \text{length } \widehat{AB} + \text{length } \widehat{BC}$.
37. The measure of an angle inscribed in a circle is one half the

measure of the central angle which subtends the same arc.